

ORIGINAL ARTICLE

Parenting pain away: Quasi-experimental study of an eHealth learning platform to evaluate acceptability, feasibility, and utilisation of parent-led pain management

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Abstract

Aim: To evaluate the impact of an eHealth educational resource about infant procedural pain management, given during the prenatal period, on feasibility, acceptability, knowledge, self-efficacy, and involvement.

Background: Routine health care requires newborns to have painful procedures (e.g., intramuscular injection). The impacts of untreated pain in neonates are widely recognised but adoption of effective procedural pain management strategies in clinical practice varies. There is clear evidence supporting the effectiveness of parent-led pain management during procedures (e.g., skin-to-skin care) and reputable resources to raise awareness among parents are warranted. Our team co-created Parenting Pain Away, a website to equip parents with evidence to assist with managing the pain of procedures and empower them to be involved.

Methods: A quasi-experimental evaluation using a pre/post intervention design with low-risk expectant parents.

Results: Of the 41 participants, before intervention exposure participants were familiar or had used skin-to-skin care ($n=33$), breastfeeding ($n=30$) and sucrose ($n=13$) as pain management. Most participants ($n=38$) desired more information on how to be involved. Providing access to Parenting Pain Away during pregnancy was supported and participants ranked the website above average using the System Usability Scale. Parenting Pain Away did not have a statistically significant influence on outcomes. Participants reported variation in clinical support with parent-led pain management.

Conclusion: A multifaceted approach is recommended to maintain infant procedural pain management.

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Implications for the Profession and Patient Care: Equipping parents with knowledge related to infant pain management using an eHealth approach satisfied their information desires. The study findings are important considerations for perinatal care providers, policy makers, and families to finally achieve adequate procedural pain management.

Reporting Method: This study used the STROBE checklist, adhering to EQUATOR guidelines.

Patient and Public Contribution: A stakeholder group (expectant parents, parents, perinatal researchers, clinicians, and administrators) was created to inform the study design and intervention.

KEYWORDS

eHealth interventions, neonatal pain management, parental involvement, prenatal education, procedural pain

BACKGROUND

In the first hours of life, newborns require tissue-breaking procedures to help support and monitor their status [1, 2]. The number of procedures varies dependent on gestational age at birth and health status. Even healthy, term-born infants (i.e., 37–40 weeks gestation) can receive up to 20 procedures during their first year of life, including vitamin K intramuscular injection as prophylaxis against bleeding [3], newborn metabolic (NBM) screening [4], bilirubin and glucose monitoring [5, 6], and immunizations [7]. These procedures are significant as untreated pain in preterm infants has been found to alter brain development, reaction to subsequent pain, as well as cognitive functioning [8–12] and in full-term infants, untreated pain has been found to heighten pain responses to subsequent pain [9, 10, 13]. There are several effective strategies for managing infant procedural pain, including parent-led pain management (PLPM) strategies, such as breastfeeding [14], skin-to-skin care (SSC) [15, 16], and sucrose [17, 18]. Application of procedural pain management and engagement of parents has varied in clinical settings [19, 20] despite available clinical practice guidelines and frequent calls to action [21–24]. Parents want to comfort their children and they highly value this role reducing their infant's pain [16, 25]. Education delivered to parents in the perinatal period has been shown to increase procedural pain management in infants [26, 27]. Yet, there is limited evidence on the effectiveness of parental education in the prenatal period [28].

Recognising the potential of anticipatory guidance to parents about procedural pain management in advance of their baby's arrival, our team developed the website Parenting Pain Away (PPA). This website was designed to target expectant parents during the prenatal period to provide comprehensive evidence about infant procedural

pain and pain management strategies [27]. PPA was developed collaboratively through consultation with stakeholders (e.g., expectant parent, clinicians, researchers, administrators), as well as a graphic designer proficient in health dissemination. Creating excellent User Experience was a priority from the onset of this project to ensure the target users of this intervention would find PPA engaging and informative, and to enhance the likelihood that the content of this website would be applied in real life. Thus, usability testing with a mix of expectant parents and parents in the postpartum period was conducted in a previous study to further refine PPA [27]. These participants ultimately felt satisfied with PPA and encouraged its integration into prenatal education [27].

AIM

Building on previous work, this study aimed to evaluate the impact of the PPA website, offered to expectant parents during the prenatal period, on feasibility, parental acceptability, knowledge, self-efficacy, and involvement.

METHODS

Study design

A quasi-experimental evaluation of PPA was conducted using a pre-/post-intervention design to answer the following questions:

1. Among parents of healthy newborns, what is the feasibility and acceptability of PPA on parent-led procedural pain management?

2. Among parents of healthy newborns, does PPA influence parental knowledge, self-efficacy, and active involvement in pain management?

Setting and sample

This study took place in the province of Prince Edward Island (PEI), located in Eastern Canada, with two hospitals providing obstetrical care for approximately 1400 births annually. Eligibility criteria included low-risk expectant parents (at least 32 weeks gestation) and/or their partners (if applicable), with internet access, proficiency in English and planning to give birth in a provincial hospital. Recruitment occurred only through online advertisements promoting the study via social media, (e.g., Twitter, Facebook, Instagram), online classifieds (e.g., Kijiji), and email to key stakeholders (e.g., Study stakeholder committee), as well as posters at relevant care areas (e.g., obstetrics clinic) and community centres (e.g., coffee shops). Interested parents were instructed to contact the lead researcher (first author) by email or text message for more information about the study and confirm eligibility. The goal was to recruit a minimum of 40 participants, consistent with previous feasibility studies [29].

Outcomes

Outcomes of interest were feasibility and acceptability of PPA, newborn pain management knowledge, parental perceived self-efficacy, and involvement in PLPM during procedures. For the purposes of this study, PLPM included any strategy where parents could be involved with their infant, such as breastfeeding, SSC, sucrose (advocating for its use), topical anaesthetics, facilitated tucking, and holding the baby when clothed. Feasibility and acceptability were measured post-intervention exposure, using a study-specific survey compiling the System Usability Scale (SUS), a standardised, reliable, and widely validated questionnaire [30, 31], and 10 additional items related to intervention delivery timing, resource content, intervention use, staff support and intentions for PLPM with future procedures. Due to its popularity and value as a tool to assess the usability of technological systems, SUS was essential to include in the feasibility and acceptability survey to also facilitate a comparison of the score derived from participants' assessment of PPA against global benchmarks [32]. Intervention usage was also gathered through website analytics. Knowledge of newborn pain management was measured pre- and post-intervention exposure using a study-specific questionnaire with items compiled from three original standardised measurements; developed

and validated by Taddio et al. [33], Harrison et al. [34], and Orr et al. [35]. Parental perceived self-efficacy was measured post-intervention exposure using the Karitane Parenting Confidence Scale (KPCS) [36], a 15-item tool that assesses the perceived self-efficacy of parents of newborns. Involvement in PLPM during procedures occurring in the hospital (i.e., vitamin K injection, NBM screening) was measured through self-report post-intervention exposure. Parental diaries were also collected to gather details related to procedures while in the hospital.

Data collection

At study enrolment, participants completed an online questionnaire to gather baseline characteristics (e.g., demographics, technology use and behaviours) and newborn pain management knowledge. Participants were then provided with instructions on how to use the PPA website and encouraged to use the website as often as they liked. Follow-up data collection began after participants gave birth with a daily prompt to complete a diary entry describing needle-related procedures conducted on their infant. One week postpartum, participants completed the final survey. All participant data were collected using surveys via REDCap (Research Electronic Data Capture), a software platform hosted by Dalhousie University [37, 38].

Data analysis

Analysis was conducted using IBM SPSS software version 29.0 and Microsoft Excel [39, 40]. Descriptive statistics and frequency were calculated to describe variables of interest. One-way analysis of variance was conducted to determine the effect of PPA on parental knowledge and perceived self-efficacy scores. Knowledge questionnaires were compared using dependent t-tests and independent t-tests to determine associations with demographics and uptake of PLPM. Sub-analyses were performed, using chi-square and binary logistic regression, to further explore possible associations with uptake of PLPM based on demographics. Any narrative responses to open-ended questions in the baseline and follow-up questionnaires and daily diaries in-hospital were analysed through content analysis and coded into broader categories [41]. All data were de-identified during analysis to reduce the risk of bias.

Conceptual framework

The Conceptual Model of Parental Involvement with Infant Pain Management by Franck and colleagues

(2012) guided this study and the development of PPA [42]. This conceptual model outlines enablers and barriers to parental involvement for procedural pain management categorised by three dimensions: Parental Beliefs; Information and Support; and Parent-Infant Proximity [42]. In alignment with this conceptual model, PPA was designed to facilitate PLPM by providing access to comprehensive information about family-centred care, infant pain management, parents capacity to comfort, and the postnatal experience. Additionally, PPA aimed to empower parents to collaborate with their healthcare providers to achieve their desired level of involvement during procedures.

Ethical considerations

This study was approved by Dalhousie University (REB #2020–5079) and Health PEI Research Ethics Boards. All participants provided informed consent following the Tri-Council Policy Statement [43].

RESULTS

Through March to October 2021, 51 participants were enrolled, of which 41 completed all aspects of the study. The sample included mothers ($n=34$, 82.3%) and fathers ($n=7$, 17.1%), predominantly with participants who identified as White, married or common law, and had completed varying levels of post-secondary education. The sample included 16 first-time parents (36.0%) and 25 parents with other children (61.0%) (Table 1).

In terms of technological preferences and behaviour, four options were presented for participants to select which devices they use to access the internet, including a desktop computer, laptop, tablet, and mobile phone. Nearly all participants ($n=40$, 97.6%) reported using their mobile device (e.g., smartphone) to access the internet with most using daily ($n=39$, 97.5%). Just over half of participants used their laptop to access the internet ($n=24$, 58.5%) but only seven participants did this daily (17.1%). All but one participant confirmed that they search for health information on the internet, with most preferring to use their mobile device ($n=35$, 85.4%). Over half of the participants identified using Google ($n=24$, 58.5%) as their first source to search, followed by medical journal websites ($n=6$, 14.6%), health websites (e.g., Mayo Clinic; $n=6$, 14.6%), or hospital/university-based websites ($n=3$, 7.3%) with one participant specifying a Facebook group (2.4%). Most participants checked information they found on the internet with formal or informal sources, with the top sources being family physicians ($n=30$, 73.2%), friends/family in

healthcare ($n=27$, 65.9%), family ($n=27$, 65.9%), friends ($n=22$, 53.7%), or a pharmacist ($n=14$, 34.1%).

Prior to the PPA intervention exposure, many participants were familiar with or used SSC ($n=33$, 80.5%) and breastfeeding ($n=30$, 73.2%) for infant procedural pain management, however fewer were familiar with or used sucrose ($n=13$, 31.7%) and one participant (2.4%) selected the “prefer not to respond” option in the questionnaire. In total, 34 participants (82.9%) had either used or knew about the effectiveness of at least one of these strategies. Sixteen participants (39.0%) reported that a healthcare provider (HCP) educated them on comfort strategies when their baby experienced pain during procedures. Twenty-one participants (51.2%) felt like they had sufficient information to provide comfort during procedures, but 38 participants (92.7%) wanted more information on how to be actively involved. Only 10 participants (24.4%) used the internet to access information about managing their infant's pain, with nine reporting it as useful (22.0%) and most searched via Google ($n=7$, 70%). Please see Table 2 for an outline of previous knowledge or use of PLPM strategies, categorised by parity.

Acceptability and feasibility

During the study period, 172 unique devices (mobile device [$n=117$, 52%]; desktop/laptop [$n=106$, 48%]) accessed PPA with a total 223 visits as captured using website analytics. Most devices were in Canada ($n=123$, 55.2%), but there were also devices that accessed the site from other countries. Self-reported frequency of website use varied with participants reporting they never used PPA ($n=14$, 35%), rarely did ($n=11$, 27.5%), or used PPA at least once every 2 weeks ($n=15$, 37.5%). The average on the SUS was 71.04 (standard deviation = 26.83), which is characterised as above average and just below “Good”.

Just over half of the participants ($n=23$, 56.1%) felt the evidence-based information provided on the website was relevant during pregnancy, while six participants disagreed (14.6%) and the remaining preferred not to respond ($n=12$, 29.3%). Participants had the opportunity to comment further on this response. Those in agreement predominantly felt it supports the preparation, with one participant sharing that “it helped you feel more comfortable in being able to handle any complications that might come with babe when it's born and is a great resource for new or experienced parents!” (first-time parent) and another acknowledging “I didn't actually use the website because I forgot about it. It was a hectic few days weeks leading up to birth. But using it after I see the value in prepping new parents for painful procedures for their newborn while in hospital” (first-time parent).

Participants also felt PPA helped increase awareness and provided “clear and concise ways to lower stress and pain for babies, not all I was fully aware of before starting to read” (parent with other children). Of those who did not feel the website was relevant during pregnancy, this was attributed to time being limited “I didn’t really get time to use it before baby came.” (parent with other children) or participants were already overwhelmed “it’s good general information but until you need it, it’s just something else to read up on when you’re already overwhelmed preparing for a newborn. Definitely a good resource but better to go over and read after baby is born.” (first-time parent).

TABLE 1 Participant characteristics.

Demographics	n (%)
Age	32 (3.43) ^a
Parent type	
Mother	34 (82.9%)
Father	7 (17.1%)
Race	
White	40 (97.6%)
Asian	1 (2.4%)
Relationship type	
Married or Common Law	35 (85.4%)
In a relationship	5 (12.2%)
Without a partner	1 (2.4%)
Education	
High School Diploma	5 (12.2%)
College Diploma	11 (26.8%)
University Degree	17 (41.5%)
Graduate Degree	7 (17.1%)
Employment	36 (87.8%)
Geographic location	
Urban	21 (51.2%)
Rural	20 (48.8%)
Parity	
Primipara	16 (39.0%)
Multipara	25 (61.0%)

^aMean (standard deviation).

When asked if the study or website could be changed to improve participants’ experience, 20 responded with no (48.8%). There were some suggestions for more prompts to interact with the resource, with one participant recommending a different delivery format: “Maybe to have an app which prompts you to use it. I forgot about this when I had my baby and had to come back after the fact to complete the surveys” (first-time parent).

Influence of the Parenting Pain Away website

Several analyses were completed to explore if PPA influenced knowledge, self-efficacy, or involvement in procedural pain management. Results from the newborn pain management knowledge questionnaire found that on average, parent knowledge scores were slightly higher at follow-up postpartum (Mean [M] = 6.88, Standard Deviation [SD] = 1.10), but no significant statistical difference was found from the baseline assessment (M = 6.61, SD = 1.14), t [40] = -1.68, p = 0.102. Postpartum knowledge scores were analysed based on the frequency use of PPA, categorised into three groups: (1) never used PPA (n = 14; 35.0%); (2) rarely used PPA (n = 11, 27.5%); (3) used PPA at least once every 2 weeks (n = 15; 37.5%). One participant was excluded from this analysis as they preferred not to specify the frequency that they visited PPA. On average postpartum knowledge scores were similar between group 1 (never used; M = 6.93, SD = 0.83), group 2 (rarely used; M = 6.91, SD = 1.38), and group 3 (used at least once per 2 weeks; 6.87, SD = 1.19), and no significant difference was found between groups, F [2, 39] = 0.011, p = 0.989. Additional analyses related to knowledge scores are presented in Tables 3 and 4.

Perceived self-efficacy scores ranged from 33 to 44 (out of a possible total of 45) with an average of 40.8 and median of 42. Most participants demonstrated confidence, with three participants (7.3%) meeting the moderate clinical range (i.e., 31–35 score) and nine (22.0%) meeting the mild clinical range (i.e., 36–39 scores) suggesting possible lower levels of confidence in parenting. Perceived

TABLE 2 Previous knowledge or use of PLPM strategies.

PLPM strategies	Total (N = 41), N (%)	First-time parents (N = 16), N (%)	Parents with other children (N = 25), N (%)
Breastfeeding	30 (73.2%)	11 (68.8%)	19 (76%)
SSC	33 (80.5%)	12 (75%)	21 (84%)
Sucrose	13 (31.7%)	2 (12.5%)	11 (44%)
Received HCP education	16 (39.0%)	3 (18.8%)	13 (52%)

Abbreviations: HCP, healthcare provider; PLPM, parent-led pain management; SSC, skin-to-skin care.

self-efficacy scores were again analysed based on the frequency of PPA use. On average, perceived self-efficacy scores varied slightly between group 1 (never used; $M=39.07$, $SD=7.14$), group 2 (rarely used; $M=41.27$, $SD=1.85$), and group 3 (used at least once per 2 weeks; $M=40.20$, $SD=3.41$), but there was no significant statistical difference between groups, $F[2, 39]=0.648$, $p=0.529$.

For parental involvement with pain management during the vitamin K injection or NBM screening, 26 participants (63.4%) were involved in at least one procedure, of which 12 (29.3%) were involved in both. Specifically with the vitamin K injection, 15 participants (36.6%) were involved, while 24 participants (58.6%) reported that they were not, and two preferred not to respond (4.9%). Twenty-three participants (56.1%) were involved with the NBM screening, and the remaining 18 participants (43.9%) were not. Thirty-five participants (85.4%) completed at least one diary entry, of which 19 participants (54.3%) reported being involved in at least one procedure during their hospital stay (i.e., not solely the vitamin K injection or NBM screening). Participants reported using the following pain management strategies: holding ($n=15$, 78.9%), skin-to-skin care ($n=13$, 68.4%), breastfeeding ($n=11$, 57.9%), sucrose ($n=10$, 52.6%), facilitated tucking ($n=2$, 10.5%), and topical anaesthetics ($n=1$, 5.3%). Of note, one participant required brief NICU care (<4h) for their baby and reported sucrose being administered for three procedures during that time. Of the six participants (14.6%) with no previous experience or awareness of PLPM, four (66.7%) used PLPM while in hospital. Ten participants (24.4%) reported that HCPs (e.g., registered nurse, laboratory technician) invited them to be involved during procedures,

however 28 (68.3%) participants were not invited, and three preferred not to respond (7.3%). Chi-square analyses were conducted, but there were no significant associations found with use of any PLPM at the vitamin K injection or NBM screening and parity, previous use of parent-led pain management, education, or use of PPA. Binary logistic regression was conducted to explore if use of any PLPM at the vitamin K injection or NBM screening could be predicted by the same set of demographic variables or use of PPA. However, no variable significantly affected the model. Many participants reported intentions to use breastfeeding ($n=29$, 70.7%) and SSC ($n=25$, 61.0%) for future procedures (e.g., 2-month vaccinations), and 16 participants (39.0%) intended to use sucrose. Of these participants, four had no previous knowledge or use of breastfeeding (13.8%) and SSC (16%), and 10 (62.5%) had no previous knowledge or use of sucrose.

Parental involvement patterns

Focusing on the three leading PLPM strategies (i.e., breastfeeding, SSC, sucrose), we asked whether parents had concerns about using the strategies during procedures. Only a few participants shared concerns related to each strategy, with four participants commenting on each breastfeeding, SSC and sucrose. The narrative data expressing their concerns were coded into three main categories: Lack of Support, Physiological Barriers, and Scepticism. The first category was Lack of Support. Participants felt their healthcare team did not engage them during procedures with one participant reporting,

TABLE 3 Association between baseline knowledge scores and parent demographics.

Parent demographics	Total, <i>n</i> (%)	Baseline knowledge score, mean (SD)	<i>p</i> ^a
<i>Education</i>	40 (97.6%) ^b		0.636 ^a
High School	5 (12.5%)	6.00 (1.58)	
College Diploma	11 (27.5%)	6.64 (1.29)	
University Degree	17 (42.5%)	6.76 (1.15)	
Graduate Degree	7 (17.5%)	6.63 (1.15)	
<i>Parity</i>	41 (100%)		0.627 ^c
Primipara	16 (39.0%)	6.5 (1.21)	
Multipara	25 (61.0%)	6.68 (1.11)	
<i>Previous knowledge/Use of PLPM strategies</i>	40 (97.6%) ^b		0.925 ^c
No previous PLPM	6 (15.0%)	6.67 (1.51)	
Previous PLPM	34 (85.0%)	6.62 (1.10)	

Abbreviations: PLPM, parent-led pain management; SD, standard deviation.

^aOne-way ANOVA.

^bParticipant excluded from analysis as they preferred not to respond to question.

^cIndependent *t*-test.

“we were not given the option to hold the baby during these procedures. They had taken the baby to the bassinet to do it.” (parent with other children) and another stating “I didn’t feel like I was allowed to breastfeed during the heel prick [NBM screening]. The person doing the heel prick was not supportive of me even holding my baby, but I did it anyway.” (parent with other children). The second category was Physiological Barriers which was specific to breastfeeding. One participant noted that breastfeeding is “not always possible with a c-section baby, as I [father] was present for the vitamin K while mother was in recovery.” (parent with other children). Another participant highlighted breastmilk supply as a barrier, sharing: “I have low milk supply and worried I wouldn’t have any milk for my babe.” (parent with other children). The final category was Scepticism related to sucrose. One participant commented “it seemed like unnecessary sugar, and I wasn’t familiar with the ingredients, so didn’t want to use it as an option when there are others available.” (parent with other children) and another participant said, “it only temporarily boost[s] dopamine in the brain, not sustainable method.” (first-time parent). One participant reported that their concerns had been alleviated, sharing: “before doing the research, I was concerned about the impact of sucrose on their ‘fresh’ digestive tract but now I have more confidence that benefits outweigh the risks with this practice.” (parent with other children).

Participants were asked to elaborate on their responses for intention to use PLPM strategies for upcoming procedures, as presented above in the influence of PPA results. Many were in favour of breastfeeding, noting that this was their practice previously, so they intended to continue or that they believed in its comforting effect, with one participant commenting that they found it “soothing for both babe and mom” (parent with other children). Reasons to not try breastfeeding were predominantly that participants were no longer using this method in their feeding plan. As for SSC, many participants supported using this strategy in the future, with some stating the caveat of using SSC only if breastfeeding was not an option. Participants expressed a desire to support their baby and noted the ease of this strategy, especially for fathers, with one participant sharing “it is easy to do, and possible for the father to do” (parent with other children). Those not planning to use SSC disclosed feeling uncomfortable with the necessary setup and positioning, one participant commenting that they are “not ok with being exposed that much in a public setting” (parent with other children). Of the 39% who intended to use sucrose, participants felt that it is a reliable option if breastfeeding or SSC cannot be used and highlighted evidence in its efficacy. For those who indicated they would not use sucrose, many simply shared a preference for breastfeeding or SSC, but some also wondered

about its availability in the clinical setting. One participant reported, “never have I seen a nurse use it at public health, so I wonder if they even have it” (parent with other children).

Participants not invited by HCPs to be involved in procedures were asked if they felt comfortable advocating to be involved. Many participants simply said yes but some reported not having the opportunity to advocate because the procedures were done without their knowledge, while they were separated from their newborn, or not supported. One participant shared “Vitamin K and glucose monitoring were done when the baby was not with me (I was being brought to my room) so I did not have a chance to speak up. Heel stick [NBM screening] was done by a lab tech who requested baby be left in her bed so [we] were only able to comfort her with touch... I feel that educating this field of professionals on the benefits of breastfeeding/comfort techniques during blood draws could be very beneficial to new babies and parents as well as the staff doing the testing” (parent with other children). Other participants felt like the transition to postpartum was too overwhelming, with one noting that “No [to advocate for involvement in procedures], it was very overwhelming after just having given birth and everything happened so fast. If the option had of been presented to me, I would have preferred to breastfeed at the time” (parent with other children).

DISCUSSION

Searching for health information on the internet is ubiquitous [44], especially in perinatal populations [45–48], as further demonstrated in this study. Yet, infant pain management strategies remain a topic that is not frequently explored [35]. Awareness or previous use of PLPM strategies was relatively high for breastfeeding and SSC in our sample but less for sucrose. These findings contrast with results from Harrison et al. [34] where 68% of their sample knew about the analgesic effect of sucrose for procedures. While all participants did not share where they learned about PLPM, under 40% had an HCP educate them about pain management. The HCPs role is paramount in successfully achieving PLPM during infant procedures [23], yet there are still challenges with engaging parents consistently. Challenges to changing this practice are partially attributed to HCPs’ level of awareness or perceived value of PLPM [23, 49]. Nearly all participants wanted more information on their role in procedural pain management and HCP support, a finding which has been found across numerous studies [16, 23, 35] and is specifically recognised as a critical component of parental involvement [42].

Parenting Pain Away analytics must be interpreted with caution as the website was accessible by others not

TABLE 4 Associations between postpartum knowledge scores and uptake of PLPM.

Uptake of PLPM ^a	Total (N = 41), n (%)	Postpartum knowledge score, me	p ^b
No PLPM	15 (36.6%)	7.00 (0.76)	0.596
PLPM	26 (63.4%)	6.81 (1.27)	

Abbreviations: PLPM, parent-led pain management; SD, standard deviation.

^aUptake of PLPM for either the vitamin K injection or newborn metabolic screening.^bIndependent *t*-test.

participating in the study. Study participants perceived PPA as roughly “good”, as per the SUS score, which exceeds benchmarks at similar development stages for educational websites [32, 50]. However, there was evidence that the information could burden expectant parents who are already overwhelmed. To address this, future iterations of PPA might consider using the nudging theory [51, 52] to provide small pieces of information at a time, or the use of trained generative artificial intelligence (e.g., ChatGPT) to offer a conversational approach to sharing information.

Despite many participants acknowledging that they rarely or never used the website, over half still felt that the information PPA offered was relevant during pregnancy. In consideration of the results and current trends in health information seeking [47, 48], mobile delivery of PLPM information during pregnancy is supported [27]. PPA could be redesigned to reduce content or be integrated with broader perinatal education endorsed by local health institutions. PLPM information could be integrated with prenatal care at various time points, such as the initial appointment with a primary prenatal care provider (e.g., nurse practitioner, family physician, midwife, OBGYN), at 34 weeks gestation (i.e., common time for transfer of care to OBGYN), and at term (37–40 weeks gestation). Integration with prenatal care provides the reminder to review parent-targeted resources (e.g., PPA), as suggested by participants.

The PPA website had no statistically significant influence on parental knowledge, perceived self-efficacy or involvement in PLPM during routine procedures. While it is not clear that participants learned about PLPM solely from PPA, our data suggests that PPA addressed an information deficit for some with no previous knowledge or use of PLPM which influenced their pain management application and planning. A resounding theme from participants' comments was the variation in engagement and support by HCPs, a common challenge also identified in related research [21, 23]. When HCPs invite parents to be involved in pain management it creates an opportunity to address potential concerns (e.g., perceived low milk supply or neurological impacts of sucrose) and empower parents by providing alternate PLPM options or discussing current evidence [21, 53]. As reported in this study, a common barrier to PLPM is procedures occurring while

the parent is not present [21]. There is evidence suggesting that HCPs prefer to complete needle-related procedures without parents being present, often as an effort to protect parents [21, 23]. An additional reason is challenges in adjusting unit routine [21]. Another common barrier, also found in this study, is the refusal of PLPM during procedures due to perceived challenges with ergonomics [21]. There are freely available educational resources targeted to HCPs addressing ergonomics to support PLPM during procedures, including a YouTube video by Harrison and colleagues [21, 54]. Institutional policies, as well as further education and support targeting HCPs remains highly warranted to mitigate barriers to PLPM.

Study limitations include the small sample which met feasibility sample requirements but did not have sufficient power to find significant differences in the statistical analyses. A subsequent study with a larger sample is required to determine effectiveness of PPA. While best efforts were made to include standardised, validated measurements for data collection, the reliability and relevancy of some measurements (e.g., KPCS not specific to pain management experience) may have impacted the findings. Furthermore, the sample was homogenous, especially in terms of education and socioeconomic status, which is not representative of the broader childbearing population. Research on the appropriate timing of PPA access during the prenatal period, its impact on underrepresented groups, and overall effectiveness on PLPM uptake is needed. However, policy changes in the study setting are crucial prior to initiating future studies about parental involvement in infant pain management.

CONCLUSION

Parenting Pain Away was recognised as a good resource for preparing for postnatal experiences. Our goal was to empower parents by directly equipping them with trustworthy PLPM information. However, there were instances where parents were not invited to be involved, challenged when they advocated for involvement, or were unaware of when procedures occurred. In this study, many parents were not optimally supported to manage their babies' pain, an omission that requires immediate attention.

A multifaceted approach is recommended for practice change, including institutional infant pain management policy adoption, integration of PLPM information into prenatal education, access to parent-targeted educational resources (e.g., PPA), and postnatal care providers who engage parents in the care of their infant.

AUTHOR CONTRIBUTIONS

Hughes: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Visualisation, Writing – original draft, Writing – review & editing. Martin-Misener, Latimer, Smit, & McGrath: Methodology, Supervision, Validation, Writing – review & editing. Campbell-Yeo: Conceptualization, Methodology, Resources, Supervision, Validation, Writing – review & editing.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

This study was approved by Dalhousie University (REB #2020–5079) and Health PEI Research Ethics Boards (reference name: B. Richardson).

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APPENDIX 1

STROBE statement—checklist of items that should be included in reports of *cohort studies*.

	Item no	Recommendation	Page no
Title and abstract	1	(a) Indicate the study's design with a commonly used term in the title or the abstract (b) Provide in the abstract an informative and balanced summary of what was done and what was found	1
<i>Introduction</i>			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	2–3
Objectives	3	State specific objectives, including any pre-specified hypotheses	3
<i>Methods</i>			
Study design	4	Present key elements of study design early in the paper	3
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	3–5
Participants	6	(a) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up (b) For matched studies, give matching criteria and number of exposed and unexposed	3–5
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	4
Data sources/measurement	8 ^a	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	4
Bias	9	Describe any efforts to address potential sources of bias	5
Study size	10	Explain how the study size was arrived at	4
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	5
Statistical methods	12	(a) Describe all statistical methods, including those used to control for confounding (b) Describe any methods used to examine subgroups and interactions (c) Explain how missing data were addressed (d) If applicable, explain how loss to follow-up was addressed (e) Describe any sensitivity analyses	5
<i>Results</i>			
Participants	13 ^a	(a) Report numbers of individuals at each stage of study—e.g., numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed (b) Give reasons for non-participation at each stage (c) Consider use of a flow diagram	6
Descriptive data	14 ^a	(a) Give characteristics of study participants (e.g., demographic, clinical, social) and information on exposures and potential confounders (b) Indicate number of participants with missing data for each variable of interest (c) Summarise follow-up time (e.g., average and total amount)	6
Outcome data	15 ^a	Report numbers of outcome events or summary measures over time	6–9
Main results	16	(a) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (e.g., 95% confidence interval). Make clear which confounders were adjusted for and why they were included (b) Report category boundaries when continuous variables were categorized	

(Continues)

APPENDIX 1 (Continued)

	Item no	Recommendation	Page no
		(c) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	6–9
Other analyses	17	Report other analyses done—e.g., analyses of subgroups and interactions, and sensitivity analyses	10–13
<i>Discussion</i>			
Key results	18	Summarise key results with reference to study objectives	13–15
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	15
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	15
Generalisability	21	Discuss the generalisability (external validity) of the study results	15
<i>Other information</i>			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	Title page

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at <http://www.plosmedicine.org/>, Annals of Internal Medicine at <http://www.annals.org/>, and Epidemiology at <http://www.epidem.com/>). Information on the STROBE Initiative is available at <http://www.strobe-statement.org>.

^aGive information separately for exposed and unexposed groups.